Derivative Rules

MATH 145 – Applied Calculus

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Definition of the Derivative Function

Recall the definition of the derivative function.

The Derivative Function

Let f be a function and x a value in the function's domain. We define the **derivative of** f, a new function called f', by the formula

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists.

We are limited in what can be accomplished from this definition alone, we would like to develop some rules to make things smoother.

Derivative of Constant Functions

Constant Functions

For any real number c, if $f(x) = c$, then $f'(x) = 0$.

Derivative of Constant Functions

Proof: If $f(x) = c$, then:

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{c - c}{h} \\ &= \lim_{h \to 0} \frac{0}{h} \\ &= 0. \end{aligned}$$

Constant Functions

Example: Find $f'(x)$ if $f(x) = 42$.

Example: Find $g'(x)$ if $g(x) = e^{110.09} + \log(16.51) - \pi^i$.

The Constant Multiple Rule

The Constant Multiple Rule

For any real number k, if $f(x)$ is a differentiable function with derivative $f'(x)$, then $\frac{d}{dx}\left[kf(x)\right] = kf'(x)$.

The Constant Multiple Rule

Proof:

$$\begin{aligned}
\frac{d}{dx}\left[kf(x)\right] &= \lim_{h\to 0}\frac{kf(x+h)-kf(x)}{h} \\
&= \lim_{h\to 0}\frac{k\left(f(x+h)-f(x)\right)}{h} \\
&= \lim_{h\to 0} k\left(\frac{f(x+h)-f(x)}{h}\right) \\
&= k\left(\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}\right) \\
&= kf'(x).
\end{aligned}$$

The Constant Multiple Rule

Example: Find $f'(x)$ if $f(x) = 3x$.

Example: Find $g'(x)$ if $g(x) = 4x^2$.

The Sum Rule

The Sum Rule

If $f(x)$ and $g(x)$ are differentiable functions with derivatives $f'(x)$ and $g'(x)$ respectively, then $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

The Sum Rule

Proof:

$$\begin{aligned}
\frac{d}{dx}\left[f(x)+g(x)\right] &= \lim_{h\to 0}\frac{(f(x+h)+g(x+h))-(f(x)+g(x))}{h} \\
&= \lim_{h\to 0}\frac{(f(x+h)-f(x))+(g(x+h)-g(x))}{h} \\
&= \lim_{h\to 0}\left[\left(\frac{f(x+h)-f(x)}{h}\right)+\left(\frac{g(x+h)-g(x)}{h}\right)\right] \\
&= \lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right)+\lim_{h\to 0}\left(\frac{g(x+h)-g(x)}{h}\right) \\
&= f'(x)+g'(x).
\end{aligned}$$

The Sum Rule

Example: Find $f'(x)$ if $f(x) = 3x + 4$.

Example: Find $g'(x)$ if $g(x) = x^2 + 4x + 8$.

The Power Rule

For any nonzero real number n, if $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

The Power Rule

Proof (for positive integer n):

$$\begin{aligned}
\frac{d}{dx}f(x) &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0}\frac{(x+h)^n - x^n}{h} \\
&= \lim_{h\to 0}\frac{\left(x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n\right) - x^n}{h} \\
&= \lim_{h\to 0}\frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n}{h} \\
&= \lim_{h\to 0}\left(nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}\right) \\
&= nx^{n-1}.
\end{aligned}$$

Using the Binomial Theorem: $(x+y)^n = \sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k}$.

The Power Rule

Example: Find $f'(x)$ if $f(x) = 9x^2 + 6x + 3$.

Example: Find $g'(x)$ if $g(x) = \dfrac{1}{9x}$.

Example: Find $\ell'(x)$ if $\ell(x) = \sqrt{4x}$.

Example: Find $q'(x)$ if $q(x) = \dfrac{1}{\sqrt[4]{x}}$.

The Product Rule

The Product Rule

If f and g are differentiable functions, then their product $P(x) = (f \cdot g)(x) = f(x) \cdot g(x)$ is also a differentiable function, and

$$P'(x) = f(x)g'(x) + g(x)f'(x).$$

I will leave out the proof here, but I can post it later. When we get to implicit differentiation and logarithms, it will become a lot easier to prove.

The Product Rule

Example: Find $f'(x)$ if $f(x) = (3x^2 - 4x)(x^4 - 1)$.

Example: Find $g'(x)$ if $g(x) = 4x^5(5x^3 - 3\sqrt{x})$.

Example: Find $\ell'(x)$ if $\ell(x) = \dfrac{1}{x}(2x^3 + 7x^6)$.

The Quotient Rule

The Quotient Rule

If f and g are differentiable functions, then their quotient $Q(x) = \frac{f(x)}{g(x)}$ is also a differentiable function for all x where $g(x) \neq 0$, where

$$Q'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{\left(g(x)\right)^2}.$$

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I like to remember this as "lo d-hi minus hi d-lo over lo squared".

The Quotient Rule

Example: Find $f'(x)$ if $f(x) = \dfrac{x^2 + 4}{x^4}$.

Example: Find $g'(x)$ if $g(x) = \dfrac{2x^3 + 7x^6}{x}$.